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From Simulated Falls to Freezing of Gait: A Controlled Study of Domain Shift in Wearable Accelerometry

Comparing CNN-1D, LSTM, GRU and a Patch Transformer under zero-shot transfer from SisFall to Daphnet

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A research report submitted as part of the requirements for the degree of
State Engineer in Computer Engineering (Génie Informatique)
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Academic year 2025–2026

June 2026

Abstract

Wearable fall detectors are trained almost exclusively on falls simulated by healthy volunteers, yet they are meant for people whose movement, and whose falls, look different. We study the most extreme version of that mismatch: a deep fall detector trained on the SisFall dataset (38 healthy young and elderly subjects) and applied, without any patient label, to freezing of gait (FoG) in the Daphnet dataset (10 patients with Parkinson's disease). The comparison is controlled: one accelerometer at the same anatomical site in both domains, rotated into a common vertical/antero-posterior/medio-lateral frame identified from the data; both domains in g at 64 Hz; fall labels centred on the detected impact; and identical seeds, mini-batches, loss, optimiser and tuning budget for a CNN-1D, an LSTM, a GRU and a patch-based Transformer. In-domain, the four models are almost indistinguishable (AUROC 0.988–0.999, recall 0.96–0.98, 0.7–1.5 false alarms per hour on unseen subjects). Under zero-shot transfer, AUROC drops to 0.49–0.63; only the Transformer's subject-level confidence interval excludes chance. At their validated thresholds the detectors stay silent on FoG (recall below 1%) while false alarms on walking rise 15- to 18-fold. A training-free freeze index ranks FoG better (AUROC 0.66) than every network. Label-free re-normalisation of the target inputs gives no consistent gain. In every learned embedding, a linear probe still separates healthy from Parkinsonian movement (AUROC ≥ 0.96), and the fall direction is nearly orthogonal to the FoG direction. Fine-tuning from fall weights is worse than training from scratch in six of eight comparisons. The transfer gap is therefore dominated by concept shift rather than covariate shift, and in-domain accuracy gives no indication of transferability.

Keywords: domain shift, fall detection, freezing of gait, Parkinson's disease, wearable sensors, Transformer, recurrent neural networks.

Acknowledgements

This work grew out of a group project carried out at ENSA Marrakech with Rokhaya Gueye and Agrhib Azar, whom I thank for their collaboration and discussions. I thank Pr. Mustapha Ameur for supervising the project. The SisFall dataset was made available by the SISTEMIC group of the Universidad de Antioquia, and the Daphnet dataset by the DAPHNet project and the Tel Aviv Sourasky Medical Center.

Contents

Abstract ii

Acknowledgements iii

1 Introduction 1
- 1.1 Problem statement . . . 2
- 1.2 Contributions . . . 2

2 Related Work 4
- 2.1 Wearable fall detection and the simulated-fall problem . . . 4
- 2.2 Freezing of gait detection . . . 4
- 2.3 Deep architectures for inertial time series . . . 5
- 2.4 Dataset shift and domain adaptation . . . 5
- 2.5 Positioning . . . 5

3 Methodology 7
- 3.1 Problem statement . . . 7
- 3.2 A common signal representation . . . 8
- 3.3 Event-level labels . . . 8
- 3.4 Robust, domain-independent scaling . . . 9
- 3.5 Architectures . . . 9
- 3.6 Training protocol and fairness . . . 10
- 3.7 Hyper-parameter search . . . 10
- 3.8 Evaluation . . . 11

4 Data and Experimental Setup 12
- 4.1 Datasets . . . 12
- 4.2 Exploratory analysis . . . 13
- 4.3 Implementation . . . 15

5 Experiments and Results **18**
5.1 Model selection on a saturated source task . . . 18
5.2 E1: in-domain fall detection . . . 19
5.3 E2: zero-shot transfer to freezing of gait . . . 19
5.4 Does normalisation fix it? . . . 21
5.5 E3: what target labels buy, and whether fall pre-training helps . . . 22
5.6 E4: covariate shift and concept shift in the representations . . . 24
5.7 Synthesis . . . 25
5.8 Limitations . . . 27

6 Conclusion **28**
6.1 Summary . . . 28
6.2 Perspectives . . . 29

References **30**

A Reproducibility **33**
A.1 Repository layout . . . 33
A.2 Running the pipeline . . . 33
A.3 Design decisions and the shortcuts they avoid . . . 34
A.4 Additional figures . . . 34

List of Tables

3.1 Hyper-parameter search spaces (six random draws per architecture). . . 11

4.1 Windows after preprocessing. SisFall training fall windows include the 0.25 s-grid copies. 12
4.2 Window descriptors: medians and Mann–Whitney rank-biserial effect size r (all $p < 10^{-10}$ unless stated). 16

5.1 Configurations selected on SisFall validation, trainable parameters, mean epochs to the selected checkpoint and training time per epoch on one NVIDIA T4 GPU. 18
5.2 In-domain fall detection on the eight SisFall test subjects (mean ± std over three seeds). Recall, specificity, F_1 and false alarms per hour (FA/h, one negative window per second of activity) use the threshold selected on validation. Best mean in bold. 19
5.3 Zero-shot transfer to Daphnet (all 17 recordings, independent domain statistics, SisFall-validated threshold). Positives are FoG windows. Best mean in bold. 20
5.4 AUROC of the seed-averaged scores with 95% subject-level bootstrap intervals. 21
5.5 Zero-shot Daphnet results under three label-free normalisations of the target inputs: AUROC (mean ± std over seeds) and, in brackets, recall / false-positive rate at the SisFall threshold. 22
5.6 Daphnet AUROC / AUPRC on held-out patients, 5-fold patient-grouped cross-validation (mean ± std over folds). FoG prevalence is 10%, the AUPRC of a random detector. 23
5.7 Effect of SisFall pre-training on Daphnet AUROC: mean difference (fine-tuned − scratch), folds improved out of five, and Wilcoxon signed-rank p (five folds, so $p \geq 0.0625$). 24

5.8 Shift between SisFall ADL and Daphnet walking, in input space and in the embedding of each SisFall-trained model: squared MMD, AUROC of a linear domain probe, under source and domain normalisation; and cosine between the fall and FoG class-difference directions. 25

A.1 Design decisions of this study and the simpler choices they replace. . . . 34

List of Figures

4.1 Preprocessing shared by both domains. Only the population and the target event differ between the two branches. 13
4.2 Left: mean acceleration per anatomical axis during SisFall walking. Right: static acceleration in the last second of fall trials, grouped by the fall direction given in the dataset documentation. 13
4.3 SisFall in the common frame. Top: slow walking. Bottom: forward fall after a slip; the shaded band is the event interval used for labelling and the dashed line the detected impact. 14
4.4 Daphnet recording with the longest freezing episode, trunk (used by the models) and ankle (for reference). The shaded band is the clinical FoG annotation. 15
4.5 Window descriptors (random subsets of up to 6,000 windows per group; outliers hidden). The freeze index is computed on the AP axis. 15
4.6 Median Welch power spectral density per group and axis (mean removed per window). Grey: locomotion band; orange: freeze band. 16
4.7 Left: hand-crafted window statistics (per-channel mean, standard deviation, percentiles and the descriptors above) embedded with t-SNE. Right: windows per split and class (log scale) and FoG prevalence per Daphnet patient. 17
4.8 Per-channel sample distributions in raw units, with the source scaler applied to both domains, and with independent per-domain scalers. . . . 17

5.1 ROC and precision-recall curves of the seed-averaged scores. Left and centre: SisFall test subjects. Right: zero-shot transfer to Daphnet. . . . 20
5.2 Confusion matrices summed over the three seeds. Top: SisFall test. Bottom: Daphnet zero-shot. Rows are normalised; counts in brackets. . 21
5.3 Seed-averaged Daphnet scores, shown as log-odds because almost all probabilities are close to zero. The dashed line is the SisFall-validated threshold. 22

5.4 Zero-shot AUROC per Daphnet patient (seed-averaged scores; S04 and S10 have no FoG). 22
5.5 Zero-shot Daphnet AUROC under three label-free normalisations (mean ± std over seeds). 23
5.6 Daphnet AUROC and AUPRC on held-out patients (mean ± std over five folds). In the zero-shot column the threshold is set on the validation patients of each fold, so its AUROC is computed on a smaller set of patients than in Table 5.3. 24
5.7 Left: AUROC of a linear probe separating SisFall ADL from Daphnet walking in each model's embedding (dotted line: input statistics). Right: cosine between the fall–ADL and FoG–walking class-mean differences in each embedding. 25
5.8 t-SNE of penultimate-layer embeddings for 800 windows per group (SisFall test subjects and Daphnet). 26
5.9 AUROC per setting and architecture (error bars: std over seeds for SisFall and zero-shot, over folds for target-label settings). Dotted line: training-free freeze index. 26

A.1 Pearson correlation between the four channels, per group. 34
A.2 Learning curves of the final runs (seed 0 opaque, seeds 1–2 transparent). 34
A.3 Hyper-parameter search. Left: validation AUPRC against model size. Right: validation AUPRC against the Daphnet AUROC that the same configuration would obtain (not used for selection). 35

Chapter 1

Introduction

Parkinson's disease is the fastest-growing neurological disorder worldwide, and falls are among its most damaging consequences (GBD 2021 Nervous System Disorders Collaborators 2024; Bloem et al. 2021). Most people with the disease fall at least once a year, and a large share fall repeatedly (Allen et al. 2013). Many of these falls are linked to *freezing of gait* (FoG): short episodes in which the feet seem glued to the floor despite the intention to walk, typically when turning, starting to walk or passing through a doorway (Nutt et al. 2011). A wearable sensor that recognised such episodes, or the falls that follow them, could trigger cueing or call for help.

Deep learning on inertial signals has made fall detection look easy. On public benchmarks, recurrent and convolutional networks routinely report sensitivities and specificities above 95% (Musci et al. 2021; Wang et al. 2019). These benchmarks, however, are built from falls simulated by young, healthy volunteers onto mattresses (Sucerquia et al. 2017; Casilari et al. 2017), and accuracy on real falls of older people is known to be much lower (Bagalà et al. 2012). Patients with Parkinson's disease add another layer: their normal gait already differs from that of healthy volunteers, and the events that matter clinically, such as freezing, do not look like the falls the models were trained on.

This report asks a direct question: does a neural network trained to detect falls on healthy subjects (SisFall) transfer, without retraining, to the detection of FoG in patients (Daphnet)? The experiment is easy to set up and easy to misread. File-level fall labels that include walking, a waist sensor compared against an ankle sensor, an interpolated target domain or a different training regime for each model would all mix confounders that have nothing to do with Parkinson's disease into the population and task differences under study. A negative result obtained under such conditions says

little about *why* transfer fails, so the protocol is designed to exclude them.

1.1 Problem statement

We study zero-shot transfer from simulated falls in healthy people to freezing of gait in Parkinson's disease, with a single trunk-worn accelerometer, under conditions designed so that the only differences between the domains are the population and the event. Within that setting we ask:

RQ1 Under identical training conditions, how do a CNN-1D, an LSTM, a GRU and a patch Transformer compare on subject-independent fall detection?

RQ2 Do these fall detectors rank FoG windows above normal Parkinsonian walking, with no patient labels at all?

RQ3 How much of the transfer gap is covariate shift that label-free normalisation could remove, and how much is concept shift?

RQ4 When a few patients are labelled, does fall pre-training help?

1.2 Contributions

- A **controlled transfer protocol** between SisFall and Daphnet: matched sensor site, units and sampling rate; an anatomical frame identified from the data; impact-centred fall labels; removal of sensor dropouts that would otherwise count as walking; and identical training conditions (seeds, mini-batch order, augmentation draws, loss, optimiser, tuning budget) for four architectures.
- A **negative result with a diagnosis**. All four models reach an AUROC of 0.988–0.999 on unseen SisFall subjects. On Daphnet they reach 0.49–0.63 and almost never fire at their validated threshold, while false alarms on walking rise 15- to 18-fold. A hand-crafted freeze index ranks FoG better than any of them without training.
- A **decomposition of the shift**. Re-normalising the target inputs changes AUROC by at most ± 0.13, with no consistent gain. A linear probe still separates healthy and Parkinsonian movement in every learned representation, and the direction that encodes "fall" is nearly orthogonal to the one that would encode "freeze". Fine-tuning from fall weights is worse than training from scratch in six of eight comparisons, and in all four at the larger label budget.
- A **reproducible pipeline**: executed notebooks, checkpoints that store their own

scaler and threshold, every table generated from result files, and a one-click GPU run.

The remainder of the report is organised as follows. Chapter 2 reviews fall and FoG detection, sequence architectures and the theory of dataset shift. Chapter 3 formalises the problem and the protocol. Chapter 4 describes the data and the exploratory analysis. Chapter 5 presents the experiments, and Chapter 6 concludes.

Chapter 2

Related Work

2.1 Wearable fall detection and the simulated-fall problem

Most public fall datasets are recorded with young volunteers who throw themselves onto mattresses while wearing an inertial sensor (Casilari et al. 2017). SisFall (Sucerquia et al. 2017) is one of the largest: 38 participants, 19 activities of daily living (ADL) and 15 fall types, recorded at the waist at 200 Hz. Its annotation is per file, so studies that need event-level labels either re-annotate it (Musci et al. 2021) or, as in the project this report revisits, assign the file label to every window. The gap between laboratory and real falls is documented: thresholds and classifiers tuned on simulated falls lose much of their sensitivity on the few real-world falls that have been recorded (Bagalà et al. 2012; Broadley et al. 2018). That literature is about the *same* event seen in a different population. The question here is harder: the event itself changes.

2.2 Freezing of gait detection

Freezing of gait (FoG) is a brief, episodic inability to produce effective stepping despite the intention to walk (Nutt et al. 2011). It is common in advanced Parkinson's disease and strongly linked to falls (Bloem et al. 2021; Allen et al. 2013). The Daphnet dataset (Bächlin et al. 2010) contains ten patients wearing three accelerometers (ankle, thigh, lower back) during walking protocols designed to provoke freezing, with video-based annotations. The classic detector is the *freeze index* of Moore et al. (2008): the ratio of power in the 3–8 Hz "freeze" band to power in the 0.5–3 Hz locomotion band, usually computed at the shank. Deep models have since been trained directly on the raw signals, including from a single waist-worn sensor (Camps et al. 2018); Pardoel et al.

(2019) review the field. All of these approaches are trained on FoG labels. We ask what happens when none are available.

2.3 Deep architectures for inertial time series

One-dimensional CNNs and recurrent networks are the standard baselines for activity recognition from wearables (Ordóñez and Roggen 2016; Wang et al. 2019; Ismail Fawaz et al. 2019). LSTMs (Hochreiter and Schmidhuber 1997) and GRUs (Cho et al. 2014) model temporal dependencies through gated recurrence; CNNs build translation-equivariant local detectors. Transformers (Vaswani et al. 2017) have more recently been adapted to time series by grouping consecutive samples into patches that act as tokens, which shortens the sequence and gives each token a local receptive field (Nie et al. 2023; Zerveas et al. 2021). Whether attention-based models generalise better under distribution shift than recurrent ones is an open empirical question for small biomedical datasets; we treat it as one.

2.4 Dataset shift and domain adaptation

Dataset shift covers any difference between the training and deployment joint distributions (Quiñonero-Candela et al. 2009; Moreno-Torres et al. 2012). It is useful to separate *covariate shift* ($p(x)$ changes, $p(y \mid x)$ does not), *label shift* ($p(y)$ changes; Lipton et al. 2018) and *concept shift* ($p(y \mid x)$ changes). Domain-adaptation theory bounds target error by source error plus a divergence between domains plus the error of the best joint hypothesis (Ben-David et al. 2010); the last term is the one that concept shift inflates and that no unsupervised alignment can reduce. Practical tools include kernel two-sample tests such as MMD (Gretton et al. 2012), adversarial feature alignment (Ganin et al. 2016; Wilson et al. 2020), and cheap normalisation-based adaptation such as re-estimating batch-norm or input statistics on target data (Li et al. 2017; Kim et al. 2022). Domain generalisation benchmarks have shown that careful empirical risk minimisation is hard to beat and that model selection must not use target labels unless reported as an oracle (Gulrajani and Lopez-Paz 2021); we follow that convention. Self-supervised pre-training is another route to transferable time-series features (Eldele et al. 2021), which we leave for future work.

2.5 Positioning

Studies that move a detector from one population or task to another usually report the drop in accuracy without separating its causes. A transfer experiment between SisFall and Daphnet is particularly easy to confound: file-level fall labels, a waist sensor

compared against an ankle sensor, an upsampled target domain or a different training regime per model would each change the outcome for reasons unrelated to Parkinson's disease. The present study removes these confounders by construction, compares recurrent, convolutional and attention-based models under identical conditions, and measures which part of the failure comes from the input statistics and which from the task itself.

Chapter 3

Methodology

3.1 Problem statement

Let $\mathcal{S}$ be the source domain (SisFall) and $\mathcal{T}$ the target domain (Daphnet). Each sample is a window $x \in \mathbb{R}^{T \times C}$ of tri-axial acceleration and a binary label y. In $\mathcal{S}$, $y = 1$ marks a window centred on a fall impact; in $\mathcal{T}$, $y = 1$ marks a window in which the patient is freezing. Both labels describe a *motor event that a monitoring system should flag*, which is the premise of the transfer question. We train f_θ on labelled samples from $\mathcal{S}$ only and evaluate it on $\mathcal{T}$ without updating θ (zero-shot transfer). Unlabelled target windows may be used to estimate input normalisation statistics, since such data are available in any deployment.

The question is not only whether f_θ transfers, but *why* it does or does not. We therefore separate three effects:

- **covariate shift**: the distribution of normal movement differs ($p_{\mathcal{S}}(x \mid y{=}0) \neq p_{\mathcal{T}}(x \mid y{=}0)$), because of age, pathology and sensor placement;
- **label shift**: the prevalence of positives differs (2.6% of SisFall test windows, 10.0% of Daphnet windows);
- **concept shift**: the positive class itself is a different phenomenon ($p_{\mathcal{S}}(x \mid y{=}1) \neq p_{\mathcal{T}}(x \mid y{=}1)$, and the direction that separates the classes in $\mathcal{S}$ need not separate them in $\mathcal{T}$).

Threshold-free metrics (AUROC) are insensitive to label shift, so a drop in AUROC points to covariate or concept shift. Normalising the target inputs with target statistics removes first- and second-order covariate shift per channel; whatever remains after that is attributed to the representation or to the concept.

3.2 A common signal representation

Every confounder that is not part of the scientific question is removed by construction (Figure 4.1).

One sensor, one site. SisFall carries two accelerometers and a gyroscope on the same waist board; Daphnet has accelerometers at the ankle, thigh and lower back. We keep the SisFall ADXL345 and the Daphnet trunk sensor. The two are both worn at the lower back/waist, so the anatomical site is matched. The second SisFall accelerometer is redundant, and the gyroscope has no counterpart in Daphnet.

Units and rate. SisFall counts are converted to g with the datasheet factor $2 \cdot 16/2^{13}$; Daphnet values (mg) are divided by 1000. SisFall is resampled from 200 to 64 Hz with a polyphase filter that includes anti-aliasing. We chose to downsample the source rather than upsample the target: interpolated samples carry no information and would have tripled the cost of every recurrent model.

Anatomical frame. Each accelerometer is rotated into a vertical (V), antero-posterior (AP), medio-lateral (ML) frame, and the magnitude $|a|$ is appended, giving $C = 4$ channels. For SisFall the axes are identified from the data instead of a figure in the dataset paper: the vertical axis carries $\approx 1\,g$ of static acceleration during walking, and after a fall the subject lies still, so gravity ends up on the AP axis for forward and backward falls and on the ML axis for lateral ones. The sign convention makes each axis read $+1\,g$ when it points upwards, so a subject lying prone reads $a_{AP} \approx -1\,g$. Daphnet's trunk axes are documented as forward, vertical and lateral.

Windows. Windows last 2 s ($T = 128$). Evaluation windows are cut on a 1 s grid (50% overlap) in both domains. In Daphnet, samples annotated 0 (outside the protocol) and sensor dropouts (trunk reading exactly $(0, 0, 0)$ mg) are removed, and windows are cut inside each contiguous valid segment only, so that no window joins two non-adjacent pieces of recording.

3.3 Event-level labels

SisFall provides one label per file. Labelling every window of a fall file as "fall" marks the walking that precedes the fall and the lying that follows it as positives, and inflates the positive class to roughly a third of all windows. We locate the impact as the maximum of $|a|$ in each fall trial (ignoring the first and last second) and define the

event interval $E = [t_p - 1\,\text{s},\ t_p + 1\,\text{s}]$. A window w is

$$y(w) = \begin{cases} 1 & \text{if } |w \cap E| \geq 1.5\,\text{s}, \\ 0 & \text{if } |w \cap E| = 0, \\ \text{discarded} & \text{otherwise}, \end{cases} \tag{3.1}$$

so a positive window always contains the impact with at least half a second of context on each side, and partially overlapping windows are left out rather than forced into either class. In the training set only, fall windows are also cut on a 0.25 s grid, which shows the network each impact at four temporal offsets; validation and test sets keep the 1 s grid. A Daphnet window is positive when at least half of its samples are annotated as freezing.

3.4 Robust, domain-independent scaling

Each channel is standardised as $(x - \text{median})/\text{IQR}$. Fall impacts reach several g while most data sit near 1 g, so mean and standard deviation would be dominated by a small number of windows; median and inter-quartile range are not. Source statistics are fitted on the SisFall training subjects only. For the target we compare three options, all label-free:

1. *source statistics*: Daphnet scaled with the SisFall training statistics (naive deployment);
2. *domain statistics*: statistics fitted on all Daphnet windows, independently of SisFall (default);
3. *subject statistics*: statistics fitted on each patient separately, a simple form of test-time adaptation.

3.5 Architectures

All four models map a window $x \in \mathbb{R}^{128\times 4}$ to one logit and expose an embedding (the input of the final linear layer) used in the shift analysis. Their sizes are tuned with the same budget (Section 3.7); the selected configurations are listed in Table 5.1.

CNN-1D. Three convolutional blocks (convolution, batch normalisation, ReLU), with channel widths $w, 2w, 4w$ and kernels narrowing with depth, max-pooling between blocks, global average pooling and dropout. Global pooling makes the detector indifferent to where in the window the event occurs.

LSTM and GRU. One or two stacked recurrent layers, optionally bidirectional. The embedding concatenates the last hidden state with the temporal mean of all hidden states, which protects against an event early in the window being partly forgotten by the time the sequence ends.

Patch Transformer. The window is split into non-overlapping patches of P samples ($P \in \{8, 16\}$); each patch of all four channels is linearly projected to d dimensions, a learnable `[CLS]` token and learned positional embeddings are added, and the sequence goes through L pre-norm encoder layers with four attention heads and GELU feed-forward blocks (Vaswani et al. 2017; Nie et al. 2023). The `[CLS]` output feeds the classifier.

3.6 Training protocol and fairness

For a given seed, every architecture sees the same training windows, in the same mini-batch order, with the same augmentation draws, because shuffling and augmentation use one seeded generator.

- Loss: binary cross-entropy with positive-class weight $n_{\text{neg}}/n_{\text{pos}}$.
- Optimiser: AdamW (Loshchilov and Hutter 2019), weight decay 10^{-4}, batch size 256, one warm-up epoch followed by cosine decay, gradient clipping at 1.0.
- At most 15 epochs; the checkpoint with the lowest (unweighted) SisFall validation cross-entropy is kept, with a patience of 4 epochs. Validation AUPRC saturates at 1.0 after two or three epochs and can no longer rank checkpoints (Section 5.1).
- Augmentation, in physical units before scaling: rotation about the vertical axis ($\pm 20°$), random ML sign flip, amplitude scaling (0.9–1.1) and Gaussian jitter (0.01 g) (Um et al. 2017). These mimic belt-mounting differences, not pathology.
- Decision threshold: the value that maximises F_2 on SisFall validation, reused unchanged on the SisFall test subjects and on Daphnet. F_2 weighs recall four times more than precision, reflecting the clinical cost of a missed event.
- Seeds 0, 1 and 2; results are reported as mean ± standard deviation.

3.7 Hyper-parameter search

Each architecture receives six configurations sampled at random from its own search space (Table 3.1), trained for at most five epochs (patience 2) on the same training subset (all fall windows and a fixed half of the non-fall windows). Configurations are

ranked by SisFall validation AUPRC rounded to three decimals, with ties broken by the validation cross-entropy. The Daphnet AUROC of every configuration is logged but *not* used for selection; it only serves to report the gain an oracle with access to target labels would obtain (Gulrajani and Lopez-Paz 2021).

Table 3.1: Hyper-parameter search spaces (six random draws per architecture).

Architecture	Search space
CNN-1D	width $w \in \{16, 32, 48\}$, kernel $\in \{5, 7, 9\}$, dropout $\in \{0.1, 0.3, 0.5\}$
LSTM / GRU	hidden $\in \{32, 64, 96\}$, layers $\in \{1, 2\}$, dropout $\in \{0.1, 0.3\}$, bidirectional $\in$ {no, yes}
Transformer	$d \in \{32, 64, 96\}$, layers $\in \{2, 3, 4\}$, patch $\in \{8, 16\}$, dropout $\in \{0.1, 0.2\}$
All	learning rate $\in \{5 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$

3.8 Evaluation

Metrics. AUROC and AUPRC (threshold-free; AUPRC is more informative under class imbalance (Saito and Rehmsmeier 2015)); at the validation threshold: recall (sensitivity), specificity, false-positive rate, precision, F_1, F_2 and balanced accuracy; and recall at 10% false-positive rate. For the seed-averaged scores, 95% confidence intervals on AUROC come from a cluster bootstrap that resamples *subjects*, since windows from one person are not independent.

Experiments.

E1 In-domain fall detection on eight unseen SisFall subjects.

E2 Zero-shot transfer to all Daphnet recordings, under the three target normalisations.

E3 Target-domain references with 5-fold patient-grouped cross-validation on Daphnet: training from scratch and fine-tuning from the SisFall weights (learning rate $\times 0.3$), with six or two training patients (the two with the most FoG windows), same training protocol as above. For fold k, the validation patients (checkpoint selection, threshold) are those of fold $k{+}1$. The patients are paired so that every test fold contains FoG: {S01, S04}, {S02, S10}, {S03, S05}, {S06, S07}, {S08, S09}.

E4 Shift analysis on the penultimate-layer embeddings of the SisFall models: squared MMD with an RBF kernel (Gretton et al. 2012), the AUROC of a linear domain probe, t-SNE (van der Maaten and Hinton 2008), and the cosine between the class-difference vectors $\Delta_{\mathcal{S}} = \mu_{\text{fall}} - \mu_{\text{ADL}}$ and $\Delta_{\mathcal{T}} = \mu_{\text{FoG}} - \mu_{\text{walk}}$. A negative or near-zero cosine means the direction that the network uses to flag falls does not point towards freezing: a direct, representation-level signature of concept shift.

Chapter 4

Data and Experimental Setup

4.1 Datasets

SisFall. The archive contains 4,505 trial files from 23 young adults (SA, 19–30 years) and 15 elderly volunteers (SE, 60–75 years). Elderly participants performed ADL only, with one exception: SE06, a judo expert, also simulated the 15 fall types. Every fall in the dataset was therefore performed by a healthy person, and all but 75 of the 1,798 fall files by someone under 31. ADL trials last 10–100 s (180 s for a few long recordings); fall trials last 15 s. No sample was missing and none exceeded the $\pm 16\,g$ range.

Daphnet. Seventeen recordings from ten patients with idiopathic Parkinson's disease (mean age 66.4 years), mostly recorded off medication to provoke freezing (Bächlin et al. 2010). The archive provides both a CSV re-export and the original UCI release; we use the original text files. Timestamps are integer milliseconds and alternate between 15 and 16 ms, i.e. 64.00 Hz on average. Two patients (S04, S10) never froze. In S10, 6.3 annotated minutes contain trunk readings of exactly $(0, 0, 0)$ mg, which is physically impossible since gravity alone produces about 1000 mg; these segments are sensor dropouts and are removed; kept, they would count as normal walking.

Table 4.1: Windows after preprocessing. SisFall training fall windows include the 0.25 s-grid copies.

Domain	Split	Subjects	Negatives	Positives	Positive (%)
SisFall	train	25	44,705	4,877	9.84
SisFall	validation	5	8,738	229	2.55
SisFall	test	8	14,173	381	2.62
Daphnet	all (zero-shot)	10	15,676	1,734	9.96

Subject split. SisFall training: SA01–SA15 and SE01–SE10 (including SE06). Validation: SA16–SA18, SE11, SE12. Test: SA19–SA23, SE13–SE15. Elderly volunteers in the validation and test sets contribute only negatives, which makes the false-alarm estimate representative of older users.

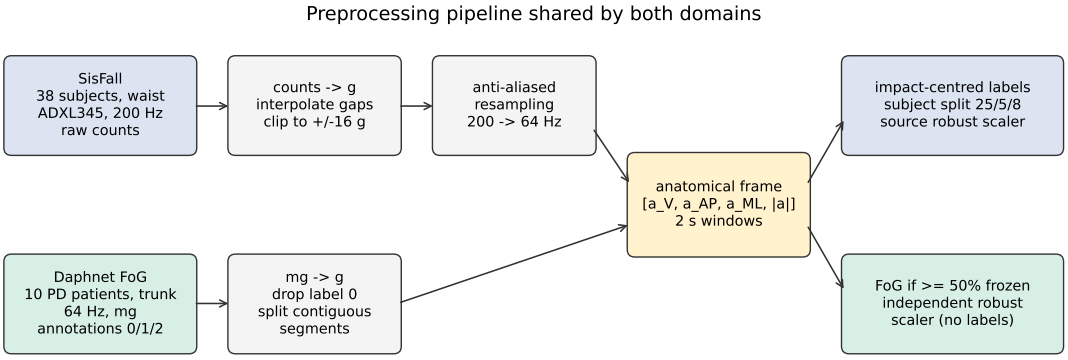


Figure 4.1: Preprocessing shared by both domains. Only the population and the target event differ between the two branches.

4.2 Exploratory analysis

Axis identification. Figure 4.2 confirms the data-driven frame: walking puts 0.97 g on the vertical axis, and the static acceleration after a fall moves to $a_{AP} = -0.60\,g$ for forward falls and $+0.83\,g$ for backward falls, i.e. prone and supine, as the sign convention predicts. The small negative AP offset during walking ($-0.2\,g$) shows that the SisFall logger was tilted on the belt by roughly 12 degrees, a reminder that "the waist" is not one orientation.

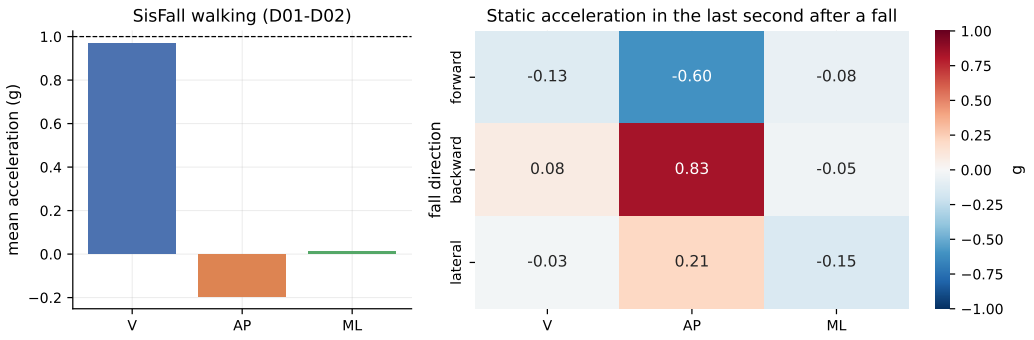


Figure 4.2: Left: mean acceleration per anatomical axis during SisFall walking. Right: static acceleration in the last second of fall trials, grouped by the fall direction given in the dataset documentation.

Signal morphology. A SisFall fall is a short, violent transient followed by rest in a new orientation (Figure 4.3). At the trunk, a Daphnet freeze looks very different: the gait rhythm becomes irregular and small, fast oscillations appear, but nothing comparable to an impact occurs (Figure 4.4). The ankle shows the trembling more clearly than the trunk, which is why the freeze index is usually computed at the shank.

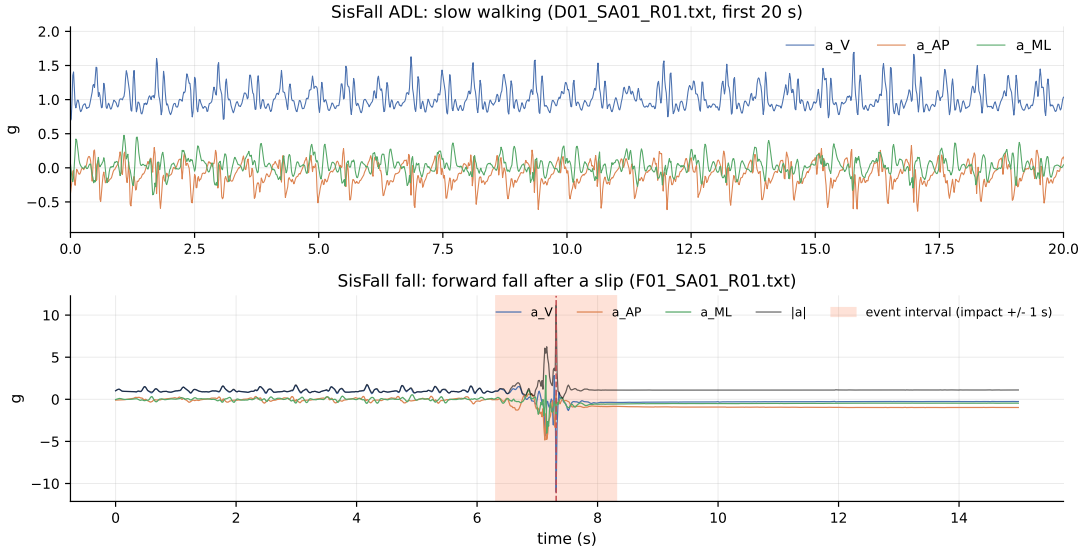


Figure 4.3: SisFall in the common frame. Top: slow walking. Bottom: forward fall after a slip; the shaded band is the event interval used for labelling and the dashed line the detected impact.

Descriptors and spectra. Figure 4.5 and Table 4.2 quantify the difference. A fall window has a median peak of 5.6 g and a $|a|$ standard deviation thirteen times that of ADL (rank-biserial $r = 0.94$ and 0.82). FoG windows differ from Parkinsonian walking much less: a factor of two in $|a|$ variability, 0.2 g in peak and +0.3 in log freeze index, all with moderate effect sizes ($r = 0.33$–0.37). The largest difference between the two *normal* classes, healthy ADL and Parkinsonian walking, is spectral: the freeze index is higher in Daphnet ($r = 0.42$) while amplitudes are almost identical. In the frequency domain (Figure 4.6), falls raise the power at every frequency by about two orders of magnitude, while FoG adds a bump between 3 and 8 Hz. The positive classes of the two domains barely overlap: in raw window statistics, falls form an isolated cluster, whereas FoG windows share most of their neighbourhoods with walking windows (Figure 4.7a). This is the first piece of evidence that the transfer problem is dominated by concept shift.

Amplitude distributions and scaling. Raw per-channel distributions of healthy ADL and Parkinsonian walking are already close (Figure 4.8, left), because both are

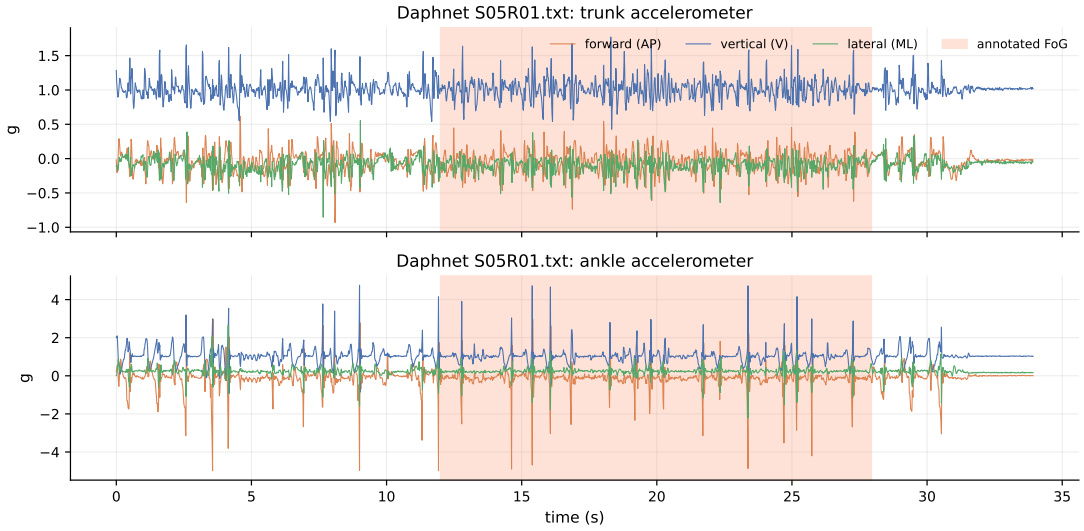


Figure 4.4: Daphnet recording with the longest freezing episode, trunk (used by the models) and ankle (for reference). The shaded band is the clinical FoG annotation.

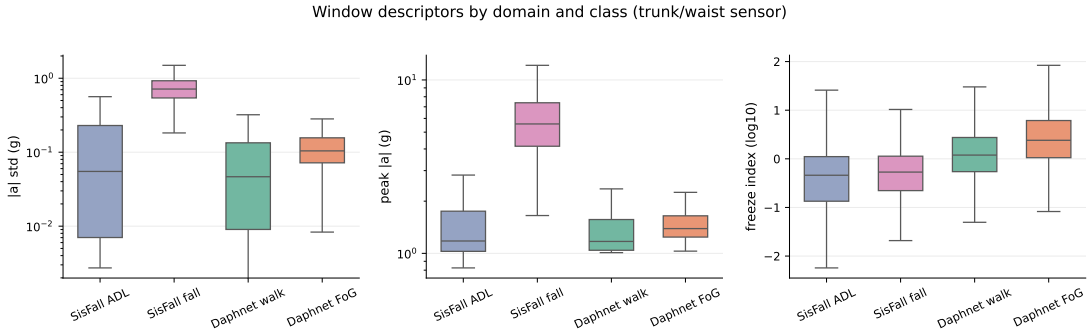


Figure 4.5: Window descriptors (random subsets of up to 6,000 windows per group; outliers hidden). The freeze index is computed on the AP axis.

dominated by upright posture. Scaling with source statistics leaves Daphnet much narrower than SisFall on the vertical and magnitude channels; scaling each domain with its own statistics widens Daphnet to unit inter-quartile range, which brings the walking distributions closer but also amplifies the small FoG oscillations. Daphnet values are stored with a resolution of about 10 mg against 3.9 mg for the ADXL345, another (minor) difference between the sensors.

4.3 Implementation

Everything is implemented in PyTorch (Paszke et al. 2019) and executed end to end in six Jupyter notebooks (data preparation, exploratory analysis, hyper-parameter search,

Table 4.2: Window descriptors: medians and Mann–Whitney rank-biserial effect size r (all $p < 10^{-10}$ unless stated).

Comparison (A vs B)	Descriptor	Median A	Median B	r
SisFall ADL vs SisFall fall	$\|a\|$ std (g)	0.055	0.714	0.82
	peak $\|a\|$ (g)	1.179	5.575	0.94
	freeze index (log10)	-0.338	-0.273	0.09 ($p = 2.5e-08$)
Daphnet walk vs Daphnet FoG	$\|a\|$ std (g)	0.047	0.104	0.37
	peak $\|a\|$ (g)	1.172	1.389	0.34
	freeze index (log10)	0.077	0.381	0.33
SisFall ADL vs Daphnet walk	$\|a\|$ std (g)	0.055	0.047	-0.04 ($p = 2.1e-05$)
	peak $\|a\|$ (g)	1.179	1.172	0.01 ($p = 5.9e-01$)
	freeze index (log10)	-0.338	0.077	0.42

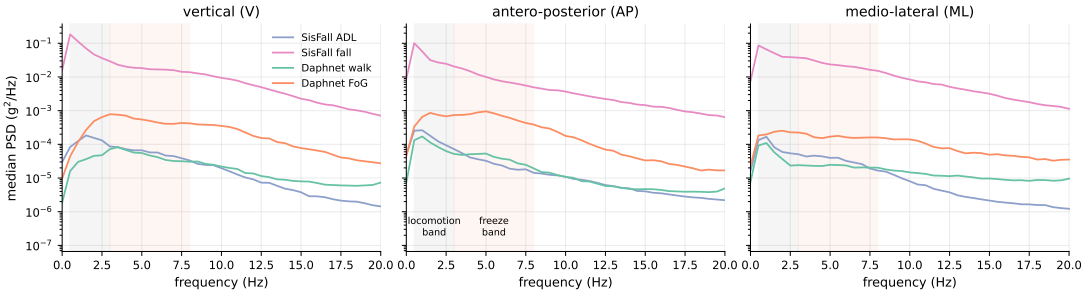
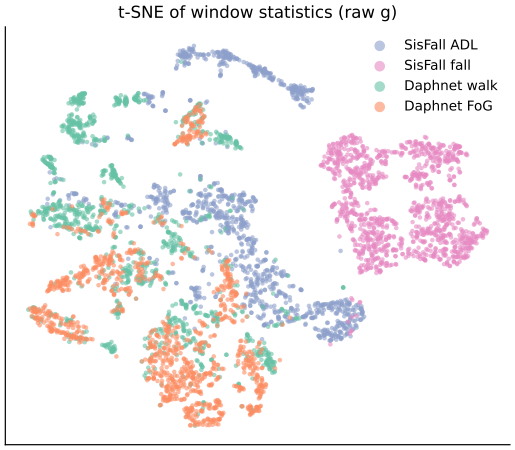
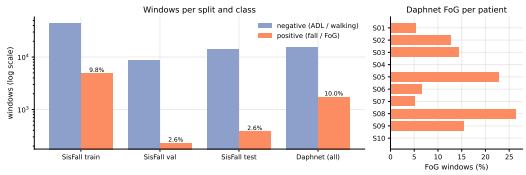


Figure 4.6: Median Welch power spectral density per group and axis (mean removed per window). Grey: locomotion band; orange: freeze band.

training and zero-shot evaluation, target adaptation, shift analysis). Shared components (models, training loop, metrics) live in a small `src` package so that the notebooks call identical code. Network training (notebooks 03–05) ran on one NVIDIA T4 GPU; preprocessing, inference from the saved checkpoints and all analyses also run on a two-core CPU, where they reproduce the reported metrics. The training cost per epoch is given in Table 5.1. Each checkpoint stores the weights together with the configuration, the scaler statistics and the validation threshold, so it is self-contained for inference.



(a) t-SNE of window statistics.



(b) Class balance per split.

Figure 4.7: Left: hand-crafted window statistics (per-channel mean, standard deviation, percentiles and the descriptors above) embedded with t-SNE. Right: windows per split and class (log scale) and FoG prevalence per Daphnet patient.

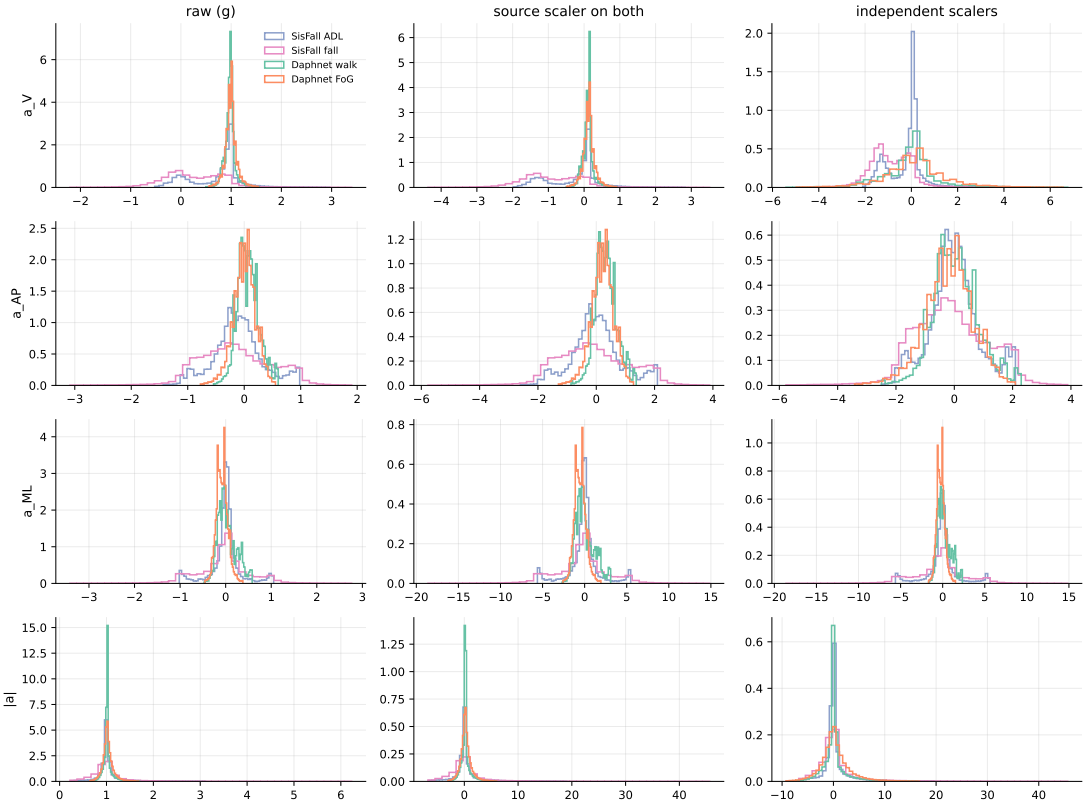


Figure 4.8: Per-channel sample distributions in raw units, with the source scaler applied to both domains, and with independent per-domain scalers.

Chapter 5

Experiments and Results

All numbers in this chapter are produced by notebooks 03–07 and stored in `results/metrics`. Unless stated otherwise, results are mean ± standard deviation over three seeds, and Daphnet inputs are normalised with independent, label-free domain statistics.

5.1 Model selection on a saturated source task

The first surprise came before any transfer experiment. With impact-centred labels, fall detection on SisFall is close to solved: 18 of the 24 configurations in the search reach a validation AUPRC of at least 0.999, and the selected models of every architecture reach 1.000 within two or three epochs (Figure A.2, Figure A.3). AUPRC therefore cannot rank configurations or checkpoints, so we break ties with the unweighted validation cross-entropy, which keeps decreasing for several more epochs. The selected models are listed in Table 5.1. The search picked a two-layer bidirectional network for both recurrent families, and the GRU ended up with the smallest model of the four (26k parameters).

Table 5.1: Configurations selected on SisFall validation, trainable parameters, mean epochs to the selected checkpoint and training time per epoch on one NVIDIA T4 GPU.

Model	Configuration	lr	Params	Best ep.	s/epoch
CNN-1D	width 32, kernel 7, dropout 0.5	3e-3	36,289	5.7	2
LSTM	hidden 64, 2 layers, bidirectional, dropout 0.3	1e-3	135,425	8.0	4
GRU	hidden 32, 2 layers, bidirectional, dropout 0.1	3e-3	26,241	9.7	2
Transformer	$d = 64$, 2 layers, patch 8, dropout 0.2	3e-3	70,401	7.3	3

Among the 18 saturated configurations, Daphnet AUROC ranges from 0.43 to 0.69 and is unrelated to validation AUPRC (Spearman $\rho = 0.23$, $p = 0.37$). It is moderately related to validation loss ($\rho = -0.49$, $p = 0.04$): better-calibrated source models transfer slightly better, which supports the tie-breaking rule but is far from a reliable selection signal. An oracle that picked the configuration with the best Daphnet AUROC would have changed the choice for the CNN only (0.54 → 0.69 in the search setting) and for the LSTM marginally (0.51 → 0.52). We do not use these oracle choices anywhere else.

5.2 E1: in-domain fall detection

Table 5.2 and the left panels of Figure 5.1 report results on the eight held-out SisFall subjects (381 fall windows, 14,173 non-fall windows, i.e. 3.9 hours of activity). The four architectures are within two points of each other on every metric. The recurrent models have the highest AUROC (0.999) and the fewest false alarms (0.7–0.8 per hour of activity). The CNN-1D has the highest recall (0.980) but a lower AUROC (0.987), caused by a handful of falls that receive very low scores; its seed-to-seed variability is also the largest (thresholds selected on validation range from 0.12 to 0.94). The Transformer misses the most falls (recall 0.959, about 16 of 381 per seed against 8 for the recurrent models). Subject-level bootstrap intervals overlap for all models (Table 5.4). In short, on the source task all four architectures are excellent, and the data do not support ranking them.

Table 5.2: In-domain fall detection on the eight SisFall test subjects (mean ± std over three seeds). Recall, specificity, F_1 and false alarms per hour (FA/h, one negative window per second of activity) use the threshold selected on validation. Best mean in bold.

Model	AUROC	AUPRC	Recall	Spec.	F_1	FA/h
CNN-1D	0.987 ± 0.003	0.980 ± 0.002	**0.980 ± 0.002**	**1.000 ± 0.000**	0.982 ± 0.006	1.5 ± 1.1
LSTM	**0.999 ± 0.000**	0.990 ± 0.002	0.977 ± 0.010	**1.000 ± 0.000**	**0.985 ± 0.005**	0.8 ± 0.0
GRU	**0.999 ± 0.000**	**0.992 ± 0.002**	0.977 ± 0.003	**1.000 ± 0.000**	**0.985 ± 0.002**	**0.7 ± 0.1**
Transformer	0.998 ± 0.001	0.984 ± 0.005	0.959 ± 0.010	**1.000 ± 0.000**	0.974 ± 0.008	0.9 ± 0.5

5.3 E2: zero-shot transfer to freezing of gait

Ranking. On Daphnet, AUROC falls to 0.49–0.63 (Table 5.3); AUPRC is 0.10–0.14 for a prevalence of 0.10. Averaging the three seeds gives 0.47 (CNN-1D), 0.58 (LSTM), 0.64 (GRU) and 0.65 (Transformer). With patients as the resampling unit, only the Transformer's 95% interval excludes chance, [0.55, 0.73] (Table 5.4). The ROC curves (Figure 5.1, right) show that whatever signal exists sits at mid-range false-positive rates, which is useless for an alarm.

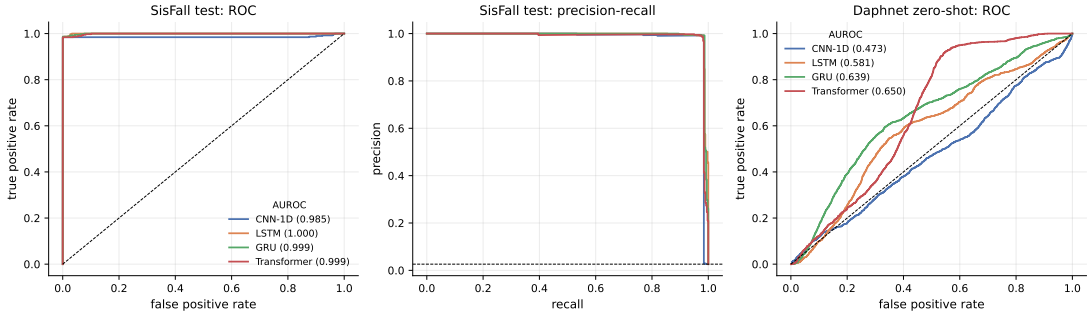


Figure 5.1: ROC and precision-recall curves of the seed-averaged scores. Left and centre: SisFall test subjects. Right: zero-shot transfer to Daphnet.

Table 5.3: Zero-shot transfer to Daphnet (all 17 recordings, independent domain statistics, SisFall-validated threshold). Positives are FoG windows. Best mean in bold.

Model	AUROC	AUPRC	Recall	Spec.	F_1	FA/h
CNN-1D	0.491 ± 0.056	0.104 ± 0.008	**0.009 ± 0.008**	0.994 ± 0.003	**0.017 ± 0.015**	23.3 ± 12.2
LSTM	0.587 ± 0.032	0.124 ± 0.021	0.001 ± 0.001	0.996 ± 0.000	0.001 ± 0.001	13.6 ± 0.9
GRU	**0.631 ± 0.042**	**0.143 ± 0.023**	0.000 ± 0.000	**0.997 ± 0.001**	0.000 ± 0.000	**11.8 ± 5.1**
Transformer	0.620 ± 0.036	0.128 ± 0.009	0.000 ± 0.000	0.995 ± 0.002	0.000 ± 0.001	17.1 ± 7.8

Operating point. At the threshold validated on SisFall, the detectors almost never fire on Daphnet. Recall is below 1% for every model, and the GRU flags none of the 1,734 FoG windows in any seed (Figure 5.2). The score distributions explain why: walking and FoG windows both receive log-odds between −10 and −4, four (CNN-1D) to ten (recurrent models, Transformer) units below the decision threshold (Figure 5.3). This silence is informative in itself. With file-level labels, the pre-fall walking of SisFall fall trials would be a positive example and irregular gait could easily look like a fall; with an ankle sensor fed where the network expects a waist sensor, the inputs alone would push the scores up. Either confounder could produce the opposite behaviour, an alarm on most Parkinsonian walking. Once the event is defined as the impact and the sensors are matched, the detectors become very specific to what they were taught.

False alarms still increase. Even so, the false-positive rate on Daphnet walking is 0.3–0.7%, which corresponds to 12–23 false alarms per hour of walking, 15 to 18 times the rate measured on healthy and elderly ADL. Covariate shift alone, without any FoG, degrades specificity by more than an order of magnitude.

A training-free reference. Two scores computed directly from each window, with no learning, do better than every network: the freeze index on the trunk AP axis reaches an AUROC of 0.663, and the standard deviation of $|a|$ reaches 0.682. At the trunk, FoG windows are on average more agitated than the walking class, which also

Table 5.4: AUROC of the seed-averaged scores with 95% subject-level bootstrap intervals.

Model	SisFall test	Daphnet zero-shot
CNN-1D	0.985 [0.961, 1.000]	0.473 [0.407, 0.546]
LSTM	1.000 [0.999, 1.000]	0.581 [0.381, 0.711]
GRU	0.999 [0.998, 1.000]	0.639 [0.458, 0.765]
Transformer	0.999 [0.997, 1.000]	0.650 [0.548, 0.733]

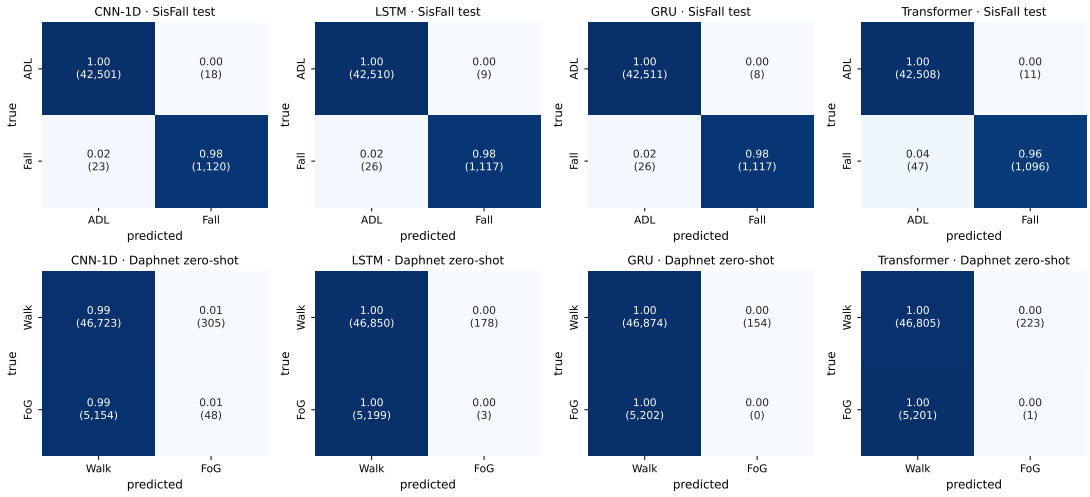


Figure 5.2: Confusion matrices summed over the three seeds. Top: SisFall test. Bottom: Daphnet zero-shot. Rows are normalised; counts in brackets.

contains standing and turning. A fall detector could in principle exploit that, but its scores carry less information about FoG than a single energy statistic.

Patients. Per-patient AUROC ranges from 0.26 to 0.85 (Figure 5.4). The recurrent models and the Transformer rank FoG above walking for S01, S02, S03 and S05 (0.60–0.85) and below it for S06 and S08 (0.27–0.49). The CNN-1D shows almost the mirror pattern. A pooled AUROC around 0.6 thus hides patient-specific behaviour in both directions.

5.4 Does normalisation fix it?

Table 5.5 and Figure 5.5 compare the three label-free normalisations of the target. No variant brings any model close to the training-free reference. Re-estimating the statistics on Daphnet helps the recurrent models and the Transformer slightly (GRU +0.06, LSTM +0.04, Transformer +0.02) and hurts the CNN-1D (−0.13); per-patient statistics fall in between. First- and second-order covariate correction at the input is

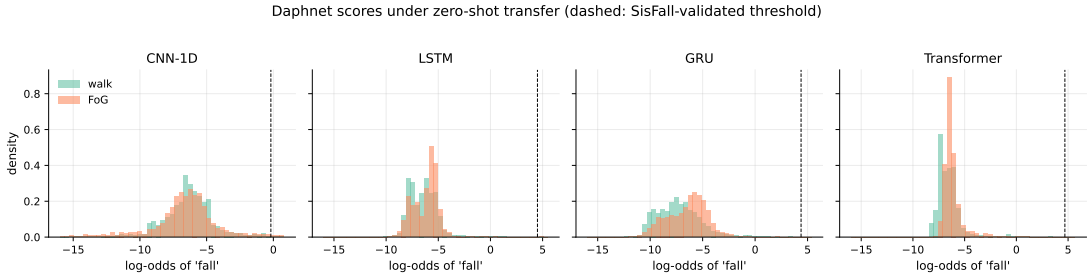


Figure 5.3: Seed-averaged Daphnet scores, shown as log-odds because almost all probabilities are close to zero. The dashed line is the SisFall-validated threshold.

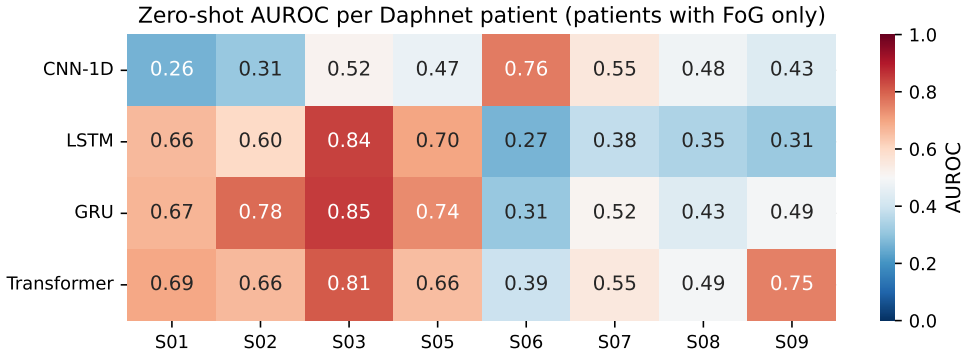


Figure 5.4: Zero-shot AUROC per Daphnet patient (seed-averaged scores; S04 and S10 have no FoG).

not the bottleneck.

Table 5.5: Zero-shot Daphnet results under three label-free normalisations of the target inputs: AUROC (mean ± std over seeds) and, in brackets, recall / false-positive rate at the SisFall threshold.

Model	Source statistics	Domain statistics	Subject statistics
CNN-1D	0.621 ± 0.064 (0.0% / 0.2%)	0.491 ± 0.056 (0.9% / 0.6%)	0.528 ± 0.042 (0.8% / 2.0%)
LSTM	0.547 ± 0.064 (0.0% / 0.1%)	0.587 ± 0.032 (0.1% / 0.4%)	0.554 ± 0.031 (0.0% / 0.4%)
GRU	0.571 ± 0.051 (0.0% / 0.1%)	0.631 ± 0.042 (0.0% / 0.3%)	0.604 ± 0.027 (0.0% / 0.3%)
Transformer	0.603 ± 0.087 (0.0% / 0.0%)	0.620 ± 0.036 (0.0% / 0.5%)	0.596 ± 0.054 (0.0% / 0.7%)

5.5 E3: what target labels buy, and whether fall pretraining helps

Table 5.6 and Figure 5.6 give the target-domain references under 5-fold patient-grouped cross-validation. Trained from scratch on six patients, the CNN-1D reaches an AUROC of 0.786, the Transformer 0.776 and the GRU 0.769, with large fold-to-fold variation (±0.15–0.19) that reflects the heterogeneity between patients. The bidirectional LSTM

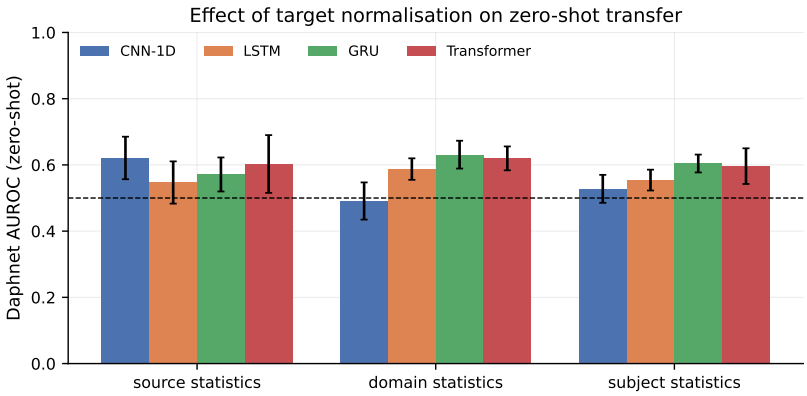


Figure 5.5: Zero-shot Daphnet AUROC under three label-free normalisations (mean ± std over seeds).

is unstable with this little data (0.608 ± 0.251). With only two training patients, the CNN-1D (0.714) and the Transformer (0.696) degrade gracefully, while the recurrent models fall to chance. This trunk-only, 2 s-window ceiling of roughly 0.78 puts the zero-shot numbers in perspective: the best zero-shot model covers about half of the distance between chance and this modest supervised ceiling, and the training-free freeze index covers more.

Table 5.6: Daphnet AUROC / AUPRC on held-out patients, 5-fold patient-grouped cross-validation (mean ± std over folds). FoG prevalence is 10%, the AUPRC of a random detector.

Setting	CNN-1D	LSTM	GRU	Transformer
Zero-shot (no target labels)	0.395±0.220 / 0.114	0.549±0.235 / 0.123	0.641±0.199 / 0.155	0.687±0.128 / 0.163
Scratch, 2 patients	0.714±0.123 / 0.238	0.351±0.256 / 0.104	0.548±0.255 / 0.195	0.696±0.153 / 0.215
Fine-tuned, 2 patients	0.617±0.144 / 0.161	0.435±0.170 / 0.089	0.588±0.167 / 0.132	0.668±0.195 / 0.179
Scratch, 6 patients	0.786±0.147 / 0.387	0.608±0.251 / 0.202	0.769±0.189 / 0.373	0.776±0.178 / 0.350
Fine-tuned, 6 patients	0.727±0.175 / 0.302	0.437±0.173 / 0.099	0.700±0.201 / 0.223	0.732±0.225 / 0.300

Fine-tuning from the SisFall weights does not help (Table 5.7). With six training patients, pre-training lowers the mean AUROC for all four architectures (by 0.04 to 0.17) and improves only one fold out of five in each case. With two patients the picture is mixed (two small gains for the recurrent models, two losses). None of the differences is significant with five folds (Wilcoxon $p \geq 0.125$), but the direction is consistent: the fall representation is a worse starting point than random initialisation. This is negative transfer.

Table 5.7: Effect of SisFall pre-training on Daphnet AUROC: mean difference (fine-tuned − scratch), folds improved out of five, and Wilcoxon signed-rank p (five folds, so $p \geq 0.0625$).

Model	Budget	Scratch	Fine-tuned	Δ	Improved	p
CNN-1D	2 patients	0.714	0.617	-0.097	1/5	0.312
CNN-1D	6 patients	0.786	0.727	-0.059	1/5	0.125
LSTM	2 patients	0.351	0.435	+0.083	3/5	0.438
LSTM	6 patients	0.608	0.437	-0.171	1/5	0.125
GRU	2 patients	0.548	0.588	+0.040	2/5	1.000
GRU	6 patients	0.769	0.700	-0.070	1/5	0.125
Transformer	2 patients	0.696	0.668	-0.028	2/5	0.625
Transformer	6 patients	0.776	0.732	-0.044	1/5	0.125

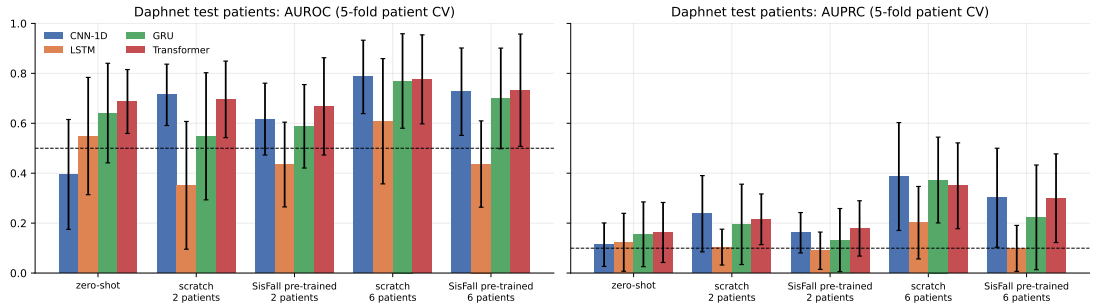


Figure 5.6: Daphnet AUROC and AUPRC on held-out patients (mean ± std over five folds). In the zero-shot column the threshold is set on the validation patients of each fold, so its AUROC is computed on a smaller set of patients than in Table 5.3.

5.6 E4: covariate shift and concept shift in the representations

Table 5.8 and Figure 5.7 examine the penultimate-layer embeddings of the SisFall models (seed 0).

Covariate shift is large and survives normalisation. A linear probe separates SisFall ADL from Daphnet walking with an AUROC of 0.96–0.995 in every space, whichever normalisation is used. Domain statistics reduce the squared MMD considerably for the LSTM (0.446 → 0.017) and the Transformer (0.128 → 0.024), but the probe accuracy hardly moves: matching moments is not the same as making the domains indistinguishable. Parkinsonian walking remains recognisable as such inside representations that were never trained to see it.

Concept shift is visible as orthogonality. In raw window statistics, the vector from walking to FoG points in the same direction as the vector from ADL to falls

(cos = +0.77): both mean "more movement". In the learned embeddings, the cosine drops to +0.18 (CNN-1D), +0.10 (LSTM), −0.03 (GRU) and +0.12 (Transformer). The networks have learned features specific to impacts and discarded the generic "agitation" direction, which is not discriminative on SisFall because jogging, stairs and stumbling share it. That direction is precisely what distinguishes FoG from walking at the trunk. The t-SNE maps (Figure 5.8) show the same picture: falls form a separate cluster, and Daphnet walking and FoG windows are interleaved with each other and partly overlap SisFall ADL.

Table 5.8: Shift between SisFall ADL and Daphnet walking, in input space and in the embedding of each SisFall-trained model: squared MMD, AUROC of a linear domain probe, under source and domain normalisation; and cosine between the fall and FoG class-difference directions.

	MMD^2		Domain probe AUROC		
Space	source st.	domain st.	source st.	domain st.	$\cos(\Delta_S, \Delta_T)$
Input statistics	0.110	0.072	0.958	0.995	+0.766
CNN-1D	0.200	0.081	0.994	0.995	+0.182
LSTM	0.446	0.017	0.989	0.975	+0.096
GRU	0.097	0.074	0.993	0.981	-0.030
Transformer	0.128	0.024	0.976	0.962	+0.122

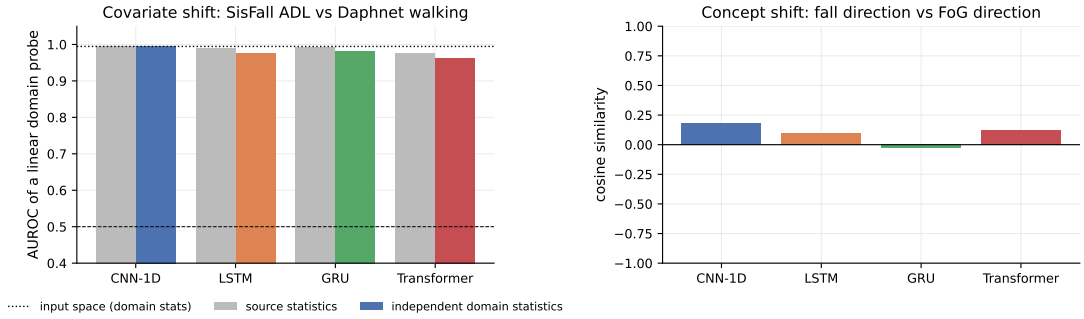


Figure 5.7: Left: AUROC of a linear probe separating SisFall ADL from Daphnet walking in each model's embedding (dotted line: input statistics). Right: cosine between the fall–ADL and FoG–walking class-mean differences in each embedding.

5.7 Synthesis

Figure 5.9 gathers every setting. Four observations summarise the study.

1. **In-domain accuracy says nothing about transfer.** All four models are near-perfect on SisFall and near chance on Daphnet. Source validation could not even rank configurations by transferability.

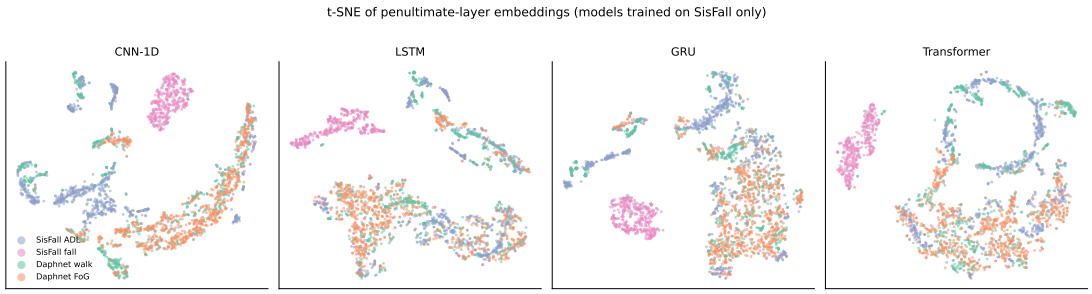


Figure 5.8: t-SNE of penultimate-layer embeddings for 800 windows per group (SisFall test subjects and Daphnet).

2. **The failure is mostly concept shift.** Normalisation removes part of the covariate shift without improving transfer; the learned fall direction is orthogonal to the FoG direction; and fall pre-training hurts fine-tuning.

3. **Architectures differ more under shift than in-domain, but not enough to matter.** The Transformer is the only model whose zero-shot interval excludes chance, and with the GRU it gives the best zero-shot ranking. The CNN-1D, worst under zero-shot transfer, is the best model once a few patients are labelled. None of these differences survives the patient-level uncertainty.

4. **Physics beats transfer.** A 2008 hand-crafted freeze index, or simply the window's energy, ranks FoG better than any fall detector, without a single label.

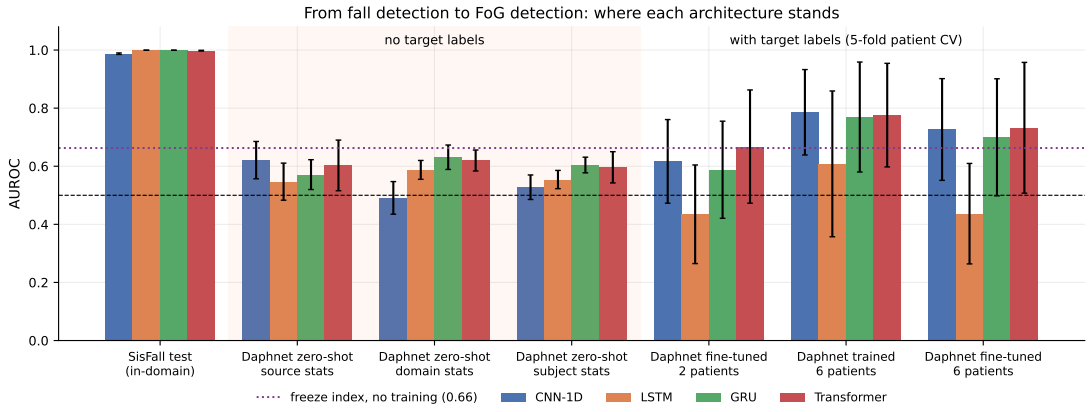


Figure 5.9: AUROC per setting and architecture (error bars: std over seeds for SisFall and zero-shot, over folds for target-label settings). Dotted line: training-free freeze index.

5.8 Limitations

Daphnet contains ten patients, two of whom never freeze, so every target-domain estimate carries wide intervals; the subject bootstrap makes that explicit, but conclusions about architectures remain tentative. The trunk sensor is the anatomically fair choice for comparison with SisFall, but it is not the best place to detect FoG; the ceiling reported in E3 should not be read as the state of the art for Daphnet, where shank sensors and longer windows are standard. Fall impacts were located automatically from the signal magnitude, which is reliable for clear impacts but may misplace the event in a few cushioned falls. SisFall falls were performed on mattresses by healthy volunteers, so even in-domain results are optimistic for real falls (Bagalà et al. 2012). Training ran on a GPU, whose recurrent kernels are not bit-exact; re-running inference from the saved checkpoints on a CPU reproduced every reported metric to four decimals.

Chapter 6

Conclusion

6.1 Summary

This study asked whether a deep fall detector trained on healthy volunteers can recognise freezing of gait in people with Parkinson's disease, and why it cannot. With the confounders removed by construction, the answers to the four research questions are as follows.

RQ1. On unseen SisFall subjects, CNN-1D, LSTM, GRU and patch Transformer are all excellent (AUROC 0.988–0.999, 0.7–1.5 false alarms per hour). The recurrent models are marginally ahead, but no ranking is supported by the subject-level uncertainty. With impact-centred labels, simulated fall detection is essentially a solved problem, so it is a poor benchmark for choosing an architecture.

RQ2. Zero-shot, the detectors rank FoG only slightly above walking (AUROC 0.49–0.63; 0.65 for the Transformer seed ensemble, the only interval that excludes chance), and at their validated operating point they detect essentially no freezing episode while raising 12–23 false alarms per hour of Parkinsonian walking. A single-line freeze index outperforms all of them.

RQ3. The gap is mostly concept shift. Label-free normalisation reduces the distance between domains without making them indistinguishable and without improving transfer, and the direction that the networks use to flag falls is nearly orthogonal to the one that separates FoG from walking.

RQ4. When patients are labelled, fall pre-training is a worse initialisation than random weights. Trained from scratch on six patients with a trunk sensor, the models reach an AUROC of about 0.78, with the CNN-1D and the Transformer the most data-efficient.

The practical lesson is simple: a monitoring system for Parkinson's disease cannot inherit its notion of "abnormal" from simulated falls. A model that is near-perfect on the source domain can be blind to the target event, and nothing measured on the source domain warns about it.

6.2 Perspectives

Event-agnostic pre-training. Fall supervision teaches impact detectors. Self-supervised objectives on large unlabelled wearable corpora (contrastive or masked reconstruction, Eldele et al. 2021) learn representations of movement itself, including the rhythmic structure that FoG disrupts, and are a better candidate for transfer.

Physics as an inductive bias. The freeze index beat every network without training. Feeding spectral features in the locomotion and freeze bands to the models, or constraining the first layers to behave like band-pass filters, could combine the robustness of the physical prior with the capacity of learned models.

Personalisation. Per-patient AUROC varied from 0.26 to 0.85, and zero-shot rankings even reversed between patients. Few-shot adaptation from a handful of annotated episodes per patient, or test-time adaptation of normalisation layers (Li et al. 2017), is a natural next step.

Better targets. Daphnet is small and was recorded in a laboratory protocol designed to provoke freezing. Larger home-recorded cohorts with both FoG and real falls, and sensors at the shank as well as the trunk, would make it possible to test whether the conclusions hold outside this setting.

References

Allen, N. E., A. K. Schwarzel, and C. G. Canning (2013). "Recurrent Falls in Parkinson's Disease: A Systematic Review". In: *Parkinson's Disease* 2013, p. 906274. DOI: 10.1155/2013/906274.

Bächlin, M. et al. (2010). "Wearable Assistant for Parkinson's Disease Patients With the Freezing of Gait Symptom". In: *IEEE Transactions on Information Technology in Biomedicine* 14.2, pp. 436–446. DOI: 10.1109/TITB.2009.2036165.

Bagalà, F. et al. (2012). "Evaluation of Accelerometer-Based Fall Detection Algorithms on Real-World Falls". In: *PLoS ONE* 7.5, e37062. DOI: 10.1371/journal.pone.0037062.

Ben-David, S. et al. (2010). "A Theory of Learning from Different Domains". In: *Machine Learning* 79.1–2, pp. 151–175. DOI: 10.1007/s10994-009-5152-4.

Bloem, B. R., M. S. Okun, and C. Klein (2021). "Parkinson's Disease". In: *The Lancet* 397.10291, pp. 2284–2303. DOI: 10.1016/S0140-6736(21)00218-X.

Broadley, R. W. et al. (2018). "Methods for the Real-World Evaluation of Fall Detection Technology: A Scoping Review". In: *Sensors* 18.7, p. 2060. DOI: 10.3390/s18072060.

Camps, J. et al. (2018). "Deep Learning for Freezing of Gait Detection in Parkinson's Disease Patients in Their Homes Using a Waist-Worn Inertial Measurement Unit". In: *Knowledge-Based Systems* 139, pp. 119–131. DOI: 10.1016/j.knosys.2017.10.017.

Casilari, E., J.-A. Santoyo-Ramón, and J.-M. Cano-García (2017). "Analysis of Public Datasets for Wearable Fall Detection Systems". In: *Sensors* 17.7, p. 1513. DOI: 10.3390/s17071513.

Cho, K. et al. (2014). "Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation". In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1724–1734. DOI: 10.3115/v1/D14-1179.

Eldele, E. et al. (2021). "Time-Series Representation Learning via Temporal and Contextual Contrasting". In: *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 2352–2359. DOI: 10.24963/ijcai.2021/324.

Ganin, Y. et al. (2016). "Domain-Adversarial Training of Neural Networks". In: *Journal of Machine Learning Research* 17.59, pp. 1–35.

GBD 2021 Nervous System Disorders Collaborators (2024). "Global, Regional, and National Burden of Disorders Affecting the Nervous System, 1990–2021: A Systematic Analysis for the Global Burden of Disease Study 2021". In: *The Lancet Neurology* 23.4, pp. 344–381. DOI: 10.1016/S1474-4422(24)00038-3.

Gretton, A. et al. (2012). "A Kernel Two-Sample Test". In: *Journal of Machine Learning Research* 13, pp. 723–773.

Gulrajani, I. and D. Lopez-Paz (2021). "In Search of Lost Domain Generalization". In: *International Conference on Learning Representations (ICLR)*.

Hochreiter, S. and J. Schmidhuber (1997). "Long Short-Term Memory". In: *Neural Computation* 9.8, pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.

Ismail Fawaz, H. et al. (2019). "Deep Learning for Time Series Classification: A Review". In: *Data Mining and Knowledge Discovery* 33.4, pp. 917–963. DOI: 10.1007/s10618-019-00619-1.

Kim, T. et al. (2022). "Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift". In: *International Conference on Learning Representations (ICLR)*.

Li, Y. et al. (2017). "Revisiting Batch Normalization For Practical Domain Adaptation". In: *International Conference on Learning Representations (ICLR), Workshop Track*.

Lipton, Z. C., Y.-X. Wang, and A. J. Smola (2018). "Detecting and Correcting for Label Shift with Black Box Predictors". In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 3122–3130.

Loshchilov, I. and F. Hutter (2019). "Decoupled Weight Decay Regularization". In: *International Conference on Learning Representations (ICLR)*.

Moore, S. T., H. G. MacDougall, and W. G. Ondo (2008). "Ambulatory Monitoring of Freezing of Gait in Parkinson's Disease". In: *Journal of Neuroscience Methods* 167.2, pp. 340–348. DOI: 10.1016/j.jneumeth.2007.08.023.

Moreno-Torres, J. G. et al. (2012). "A Unifying View on Dataset Shift in Classification". In: *Pattern Recognition* 45.1, pp. 521–530. DOI: 10.1016/j.patcog.2011.06.019.

Musci, M. et al. (2021). "Online Fall Detection Using Recurrent Neural Networks on Smart Wearable Devices". In: *IEEE Transactions on Emerging Topics in Computing* 9.3, pp. 1276–1289. DOI: 10.1109/TETC.2020.3027454.

Nie, Y. et al. (2023). "A Time Series is Worth 64 Words: Long-term Forecasting with Transformers". In: *International Conference on Learning Representations (ICLR)*.

Nutt, J. G. et al. (2011). "Freezing of Gait: Moving Forward on a Mysterious Clinical Phenomenon". In: *The Lancet Neurology* 10.8, pp. 734–744. DOI: 10.1016/S1474-4422(11)70143-0.

Ordóñez, F. J. and D. Roggen (2016). "Deep Convolutional and LSTM Recurrent Neural Networks for Multimodal Wearable Activity Recognition". In: *Sensors* 16.1, p. 115. DOI: 10.3390/s16010115.

Pardoel, S. et al. (2019). "Wearable-Sensor-Based Detection and Prediction of Freezing of Gait in Parkinson's Disease: A Review". In: *Sensors* 19.23, p. 5141. DOI: 10.3390/s19235141.

Paszke, A. et al. (2019). "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: *Advances in Neural Information Processing Systems*. Vol. 32.

Quiñonero-Candela, J. et al., eds. (2009). *Dataset Shift in Machine Learning*. MIT Press.

Saito, T. and M. Rehmsmeier (2015). "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets". In: *PLoS ONE* 10.3, e0118432. DOI: 10.1371/journal.pone.0118432.

Sucerquia, A., J. D. López, and J. F. Vargas-Bonilla (2017). "SisFall: A Fall and Movement Dataset". In: *Sensors* 17.1, p. 198. DOI: 10.3390/s17010198.

Um, T. T. et al. (2017). "Data Augmentation of Wearable Sensor Data for Parkinson's Disease Monitoring Using Convolutional Neural Networks". In: *Proceedings of the 19th ACM International Conference on Multimodal Interaction (ICMI)*, pp. 216–220. DOI: 10.1145/3136755.3136817.

Van der Maaten, L. and G. Hinton (2008). "Visualizing Data using t-SNE". In: *Journal of Machine Learning Research* 9, pp. 2579–2605.

Vaswani, A. et al. (2017). "Attention Is All You Need". In: *Advances in Neural Information Processing Systems*. Vol. 30.

Wang, J. et al. (2019). "Deep Learning for Sensor-Based Activity Recognition: A Survey". In: *Pattern Recognition Letters* 119, pp. 3–11. DOI: 10.1016/j.patrec.2018.02.010.

Wilson, G., J. R. Doppa, and D. J. Cook (2020). "Multi-Source Deep Domain Adaptation with Weak Supervision for Time-Series Sensor Data". In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 1768–1778. DOI: 10.1145/3394486.3403228.

Zerveas, G. et al. (2021). "A Transformer-based Framework for Multivariate Time Series Representation Learning". In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, pp. 2114–2124. DOI: 10.1145/3447548.3467401.

Appendix A

Reproducibility

A.1 Repository layout

```
parkinson-domain-shift/
  data/raw/archives/     SisFall_dataset.zip, Daphnet_dataset.zip (untouched)
  data/processed/        windows in g, metadata, scalers, PyTorch tensors
  notebooks/             01-07, executed, with outputs
  src/                   models, training loop, metrics, transforms
  models/<arch>/seed<k>.pt   checkpoints with config, scaler and threshold
  results/figures/       every figure of this report (PNG and PDF)
  results/metrics/       every table of this report (CSV/JSON)
  results/confusion_matrices/
  paper/                 this report and the poster (LaTeX + PDF)
```

A.2 Running the pipeline

The notebooks are executed in order. Notebooks 01, 02, 06 and 07 take a few minutes; 03–05 train neural networks and are the only expensive steps. Notebooks 03 and 05 write their results row by row and skip finished runs when restarted, and notebook 04 reloads existing checkpoints, so an interrupted run can be resumed. The training notebooks were executed on a Colab T4 GPU with `colab/RUN_ON_COLAB.ipynb`; notebooks 04, 06 and 07 were then re-executed on CPU from the saved checkpoints (inference only), which reproduced all metrics to four decimals.

Aspect	Common shortcut	This study
Fall labels	every window of a fall file	windows centred on the impact (± 0.5 s), ambiguous ones dropped
Sensors	whatever channels are available in each dataset	one waist/trunk accelerometer in both domains, in a V/AP/ML frame
Sampling rate	target upsampled to the source rate	source downsampled to 64 Hz (anti-aliased)
Units	raw counts / mg, then z-scored	g in both domains
Target windows	across concatenated recordings	inside contiguous valid segments; dropouts removed
Scaling	mean / standard deviation	robust (median/IQR), three label-free target variants
Training	tuned separately for each model	identical loss, optimiser, batches, seeds and budget
Threshold	fixed by hand	F_2-optimal on source validation

Table A.1: Design decisions of this study and the simpler choices they replace.

A.3 Design decisions and the shortcuts they avoid

A.4 Additional figures

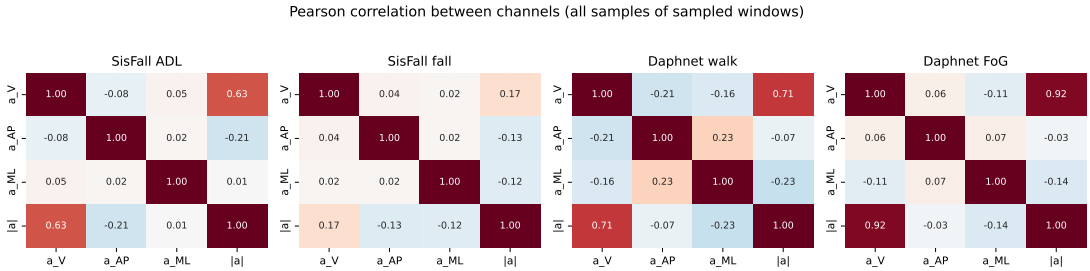


Figure A.1: Pearson correlation between the four channels, per group.

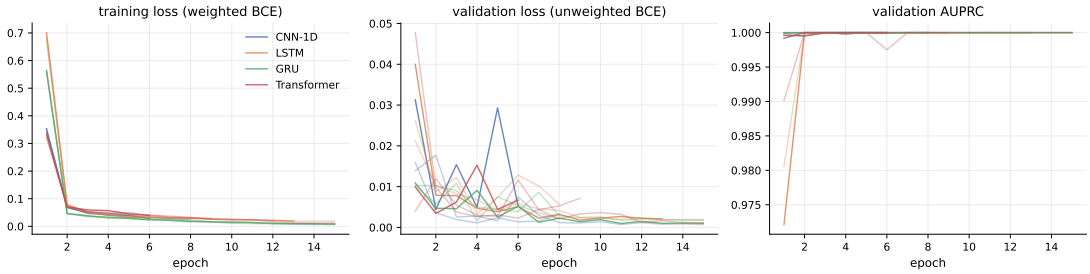


Figure A.2: Learning curves of the final runs (seed 0 opaque, seeds 1–2 transparent).

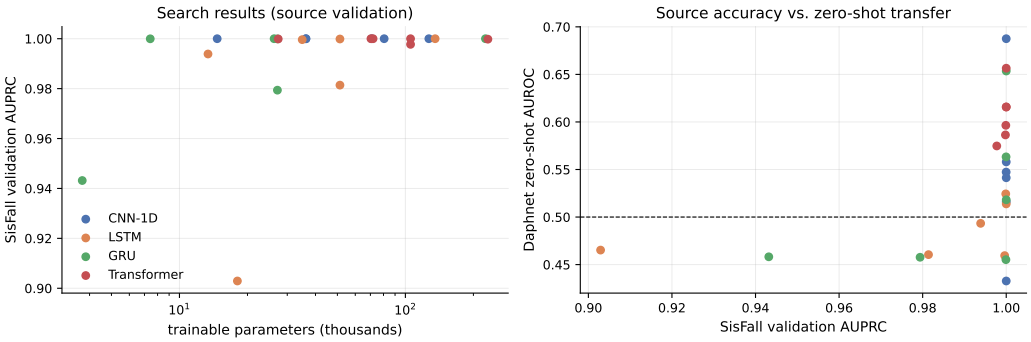


Figure A.3: Hyper-parameter search. Left: validation AUPRC against model size. Right: validation AUPRC against the Daphnet AUROC that the same configuration would obtain (not used for selection).